



DSA MARATHON

Pattern Printing — Questions & Answers

With Complete C++ Solutions

Question 1: Print the Following Pattern

Input

n = 4

Output

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

Solution

```
C++
#include <iostream>
using namespace std;
int main()
{
    // Declare variable
    int n;
    // Take input
    cin >> n;
    // Print the pattern
    for(int i = 1; i <= n; i++)
    {
        for(int j = 1; j <= n; j++)
        {
            cout << i << " ";
        }
        cout << endl;
    }
    return 0;
}
```

Question 2: Print the Following Pattern

Input

n = 4

Output

```
1 2 3 4
1 2 3
1 2
1
```

Solution

C++

```
#include <iostream>
using namespace std;
int main()
{
    // Declare variable
    int n;
    // Take input
    cin >> n;
    // Print the pattern
    for(int i = 1; i <= n; i++)
    {
        for(int j = 1; j <= n - i + 1; j++)
        {
            cout << j << " ";
        }
        cout << endl;
    }
    return 0;
}
```

Question 3: Print the Following Pattern

Input

n = 4

Output

```
A
A B
A B C
A B C D
```

Solution

C++

```
#include <iostream>
```

```
using namespace std;
int main()
{
    // Declare variable
    int n;
    // Take input
    cin >> n;
    // Print the pattern
    for(int i = 1; i <= n; i++)
    {
        for(int j = 1; j <= i; j++)
        {
            cout << char('A' + j - 1) << " ";
        }
        cout << endl;
    }
    return 0;
}
```

Question 4: Print the Following Pattern

Input

n = 5

Output

```
1
A B
1 2 3
A B C D
1 2 3 4 5
```

Solution

C++

```
#include <iostream>
using namespace std;
int main()
{
    // Declare variable
    int n;
    // Take input
    cin >> n;
    // Print the pattern
    for(int i = 1; i <= n; i++)
    {
        if(i % 2 == 1)
        {
```

```

        for(int j = 1; j <= i; j++)
        {
            cout << j << " ";
        }
    }
    else
    {
        for(int j = 1; j <= i; j++)
        {
            cout << char('A' + j - 1) << " ";
        }
    }
    cout << endl;
}
return 0;
}

```

Question 5: Print the Following Pattern

Input

n = 4

Output

```

1
1 2
1 2 3
1 2 3 4

```

Solution

C++

```

#include <iostream>
using namespace std;
int main()
{
    // Declare variable
    int n;
    // Take input
    cin >> n;
    // Print the pattern
    for(int i = 0; i < n; i++)
    {
        // Print leading spaces
        for(int j = 0; j < n - i - 1; j++)
        {
            cout << " ";
        }
    }
}

```

```
// Print numbers
for(int j = 1; j <= i + 1; j++)
{
    cout << j << " ";
}
cout << endl;
}
return 0;
}
```

End of Pattern Printing Q&A Set

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